Humanized PD-1 Mouse

Strain Name: C57BL/6-*Pdcd1* $^{em1(hPDCD1)/Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00015

Programmed cell death protein 1, also known as PD-1 or CD279, is a cell surface receptor on activated T cells. PD-1 is an important immune checkpoint molecule that negatively modulates T cell responses upon the binding of its ligand, PD-L1. Increasing evidence indicates that the PD-L1 expression on the surface of tumor cells is up-regulated in tumor micro-environment. The binding of PD-L1 to PD-1 on activated T cells results in an apoptosis or immune disability of tumor antigen-specific T cells, thereby suppressing anti-tumor immune responses. The blockade of PD-L1 binding to PD-1 reverses T cell exhaustion and thus strengthens anti-tumor activity, which has become a classic method for enabling tumor immunotherapy.

Construction strategy

On the C57BL/6 background, the full-length coding sequence of human PDCD-1 gene was placed immediately downstream of the start codon of the mouse endogenous Pdcd1, followed by a poly(A) element. This guarantees an exclusive expression of human PD-1 in the humanized PD-1 mice.

Validation data

- Data from flow cytometry (FACS) analysis

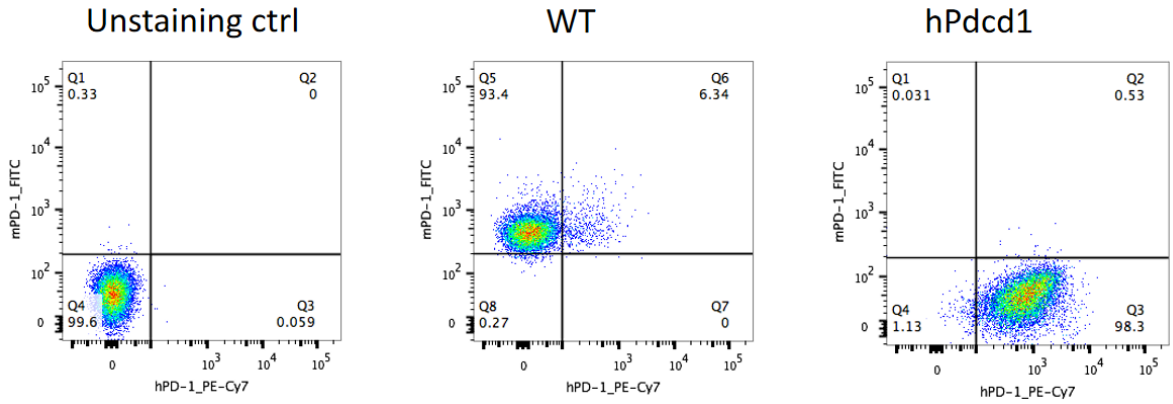


Figure 1. Expression of PD-1 in the activated spleen lymphocytes of humanized PD-1 Homozygous mice is detected by FACS.

- In vivo validation in a MC38 tumor-bearing model of humanized PD-1 mice

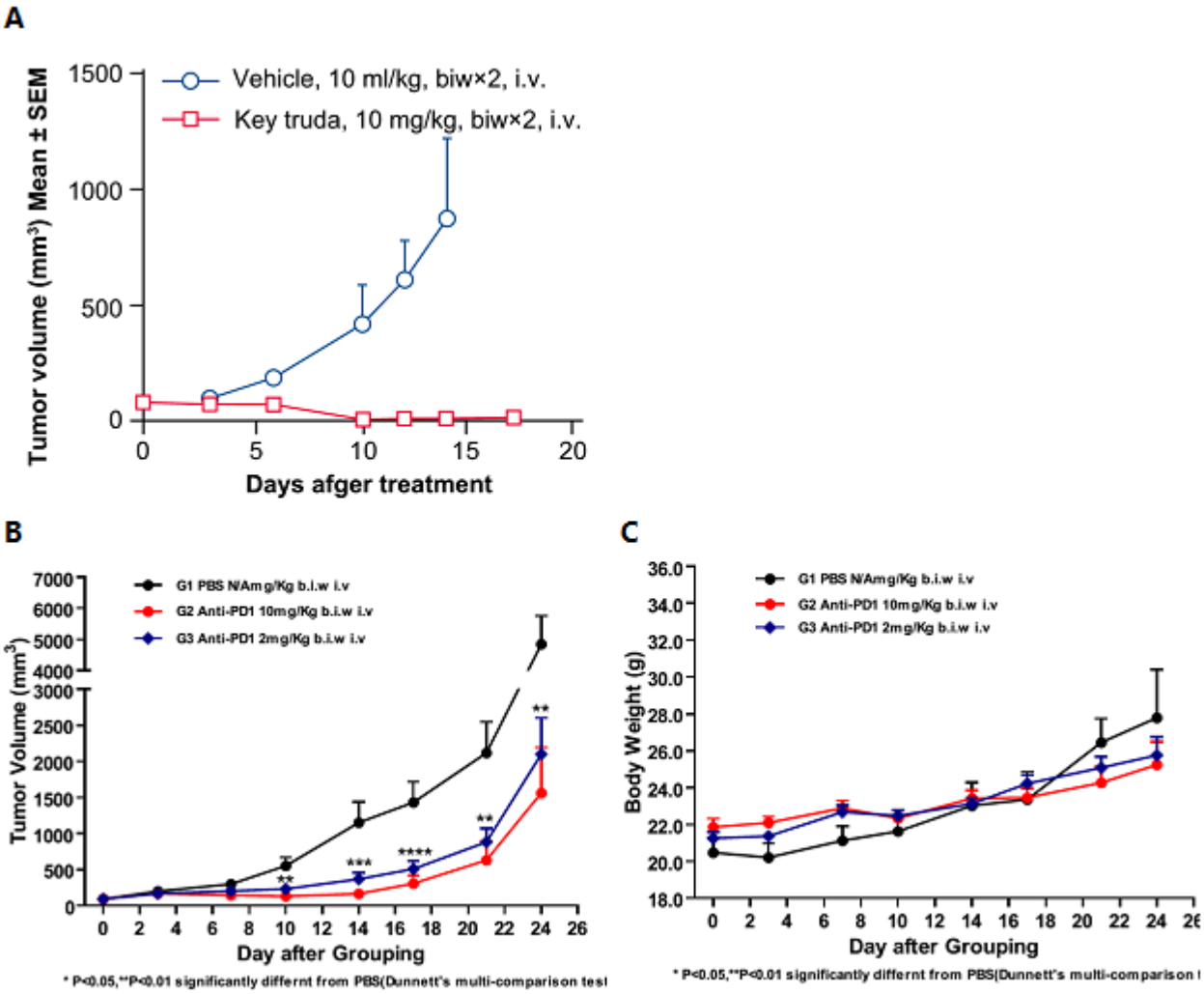


Figure 2. In vivo anti-tumor effect of an anti-human PD-1 antibody in a humanized mouse model of PD-1. Anti-human-PD-1 drugs significantly inhibited the growth of MC38 tumors in PD-1 mice, demonstrating that the humanized PD-1 mice can be used to assess the anti-human PD-1 antibody.

A. Mean volume ± SEM of tumor tissues (completed in cooperation with Genscript).

In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized PD-1 mice. Homozygous humanized PD-1 mice were inoculated with MC38 colon cancer cells. After the tumors grew to 100 mm3, the animals were randomly assigned into a control group and a treatment group (n=8). The drug was given twice a week for a total of 4 administrations. The results showed that Keytruda, a drug targeting human PD-1, exerted a very significant anti-tumor effect ($p<0.001$), demonstrating that the humanized PD-1 mice are a good in vivo model for validating the efficacy of antibodies targeting human PD-1.

B. Mean volume ± SEM of tumor tissues. C. Mean body weight ± SEM of mice (data were obtained in cooperation with PharmaLegacy).

In vivo dose validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized PD-1 mice. Homozygous humanized PD-1 mice were inoculated with MC38 colon cancer cells. After the tumors grew to about 90 mm3, the animals were randomly assigned into a control group and a treatment group (n=9). The results showed that the antibodies targeting human PD-1 showed a very significant antitumor effect ($p<0.001$), and such antitumor effect is dose-dependent.

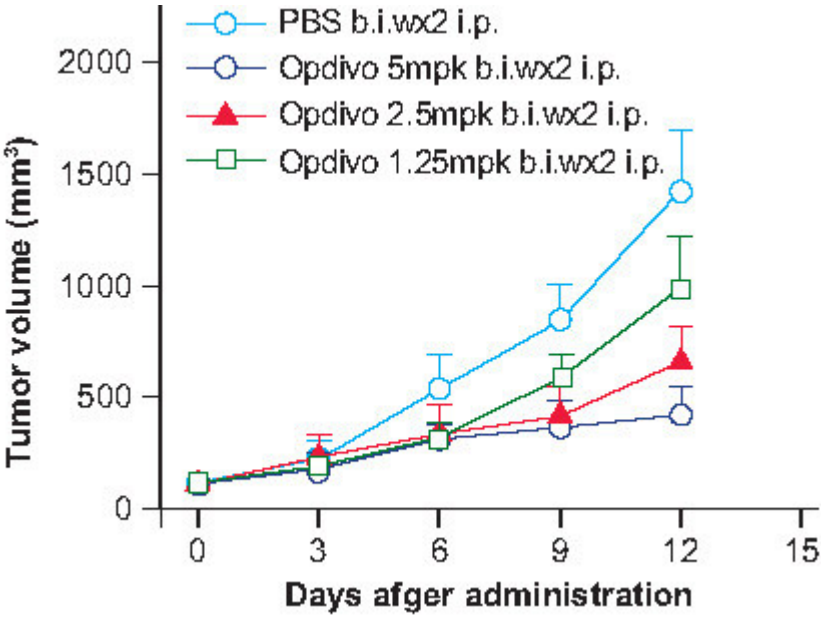


Figure 3. In vivo dose validation in a MC38 tumor-bearing model of humanized PD-1 mice. Homozygous humanized PD-1 mice were inoculated with MC38 colon cancer cells. After the tumors grew to 100 mm3, the animals were randomly assigned into a control group and a treatment group (n=8). The drug was given twice a week for a total of 4 administrations.

Case Study

As one of the first five anti-PD-1 drugs with BLAs accepted by CFDA, Sintilimab has shown potent T cell stimulating activity and significant anti-tumor efficacy. (read more)

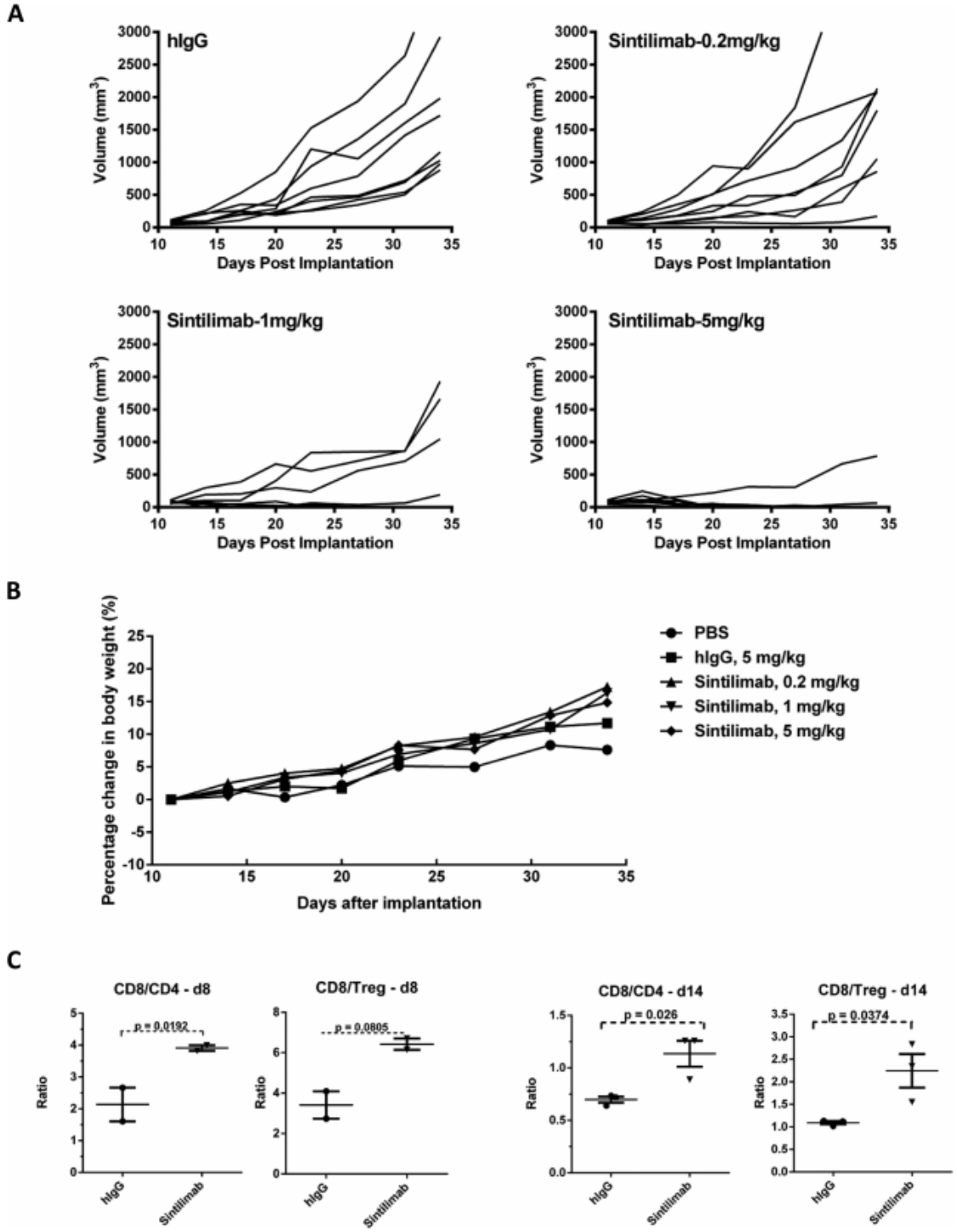


Figure 4. In vivo hPD-1 knock-in mouse model to test anti-tumor efficacy of Sintilimab. (A) Tumor growth inhibition (TGI) of MC38 tumors in hPD-1 knock-in mice of individual animals treated with different doses of Sintilimab. (B) Effect of Sintilimab on percentage changes in mouse body weight(mean). (C) Changes in ratios of tumor infiltrating CD4+, CD8+ and Treg cells. For d8: IgG (n = 2); Sintilimab (n = 2). For d14, n = 3 for all groups. P values were calculated using a two-tailed t-test method.

Name: BALB/c-*Pdcd1* $^{em1(hPDCD1)Smoc}$ Strain Background: BALB/c Cat. NO. : IT-HU-190079

Construction strategy

The BALB/c-Pdcd1 $^{em1(hPDCD1)Smoc}$ (abbreviated as BALB/c-hPD1) mouse genetic stock was developed by first crossing BALB/c females with B6-hPD1 males.

Validation data

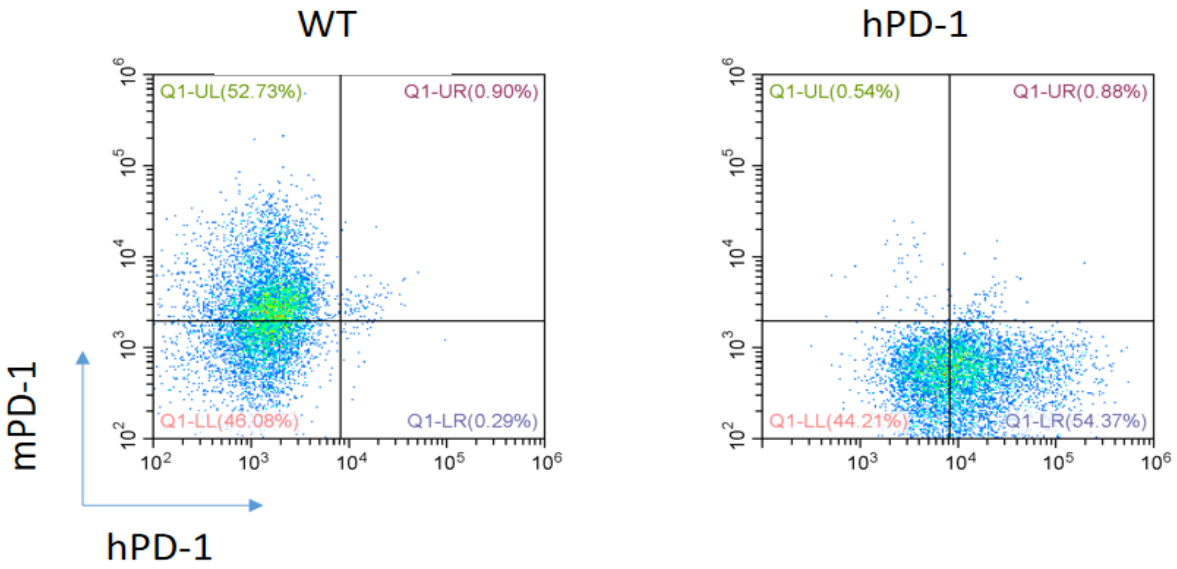
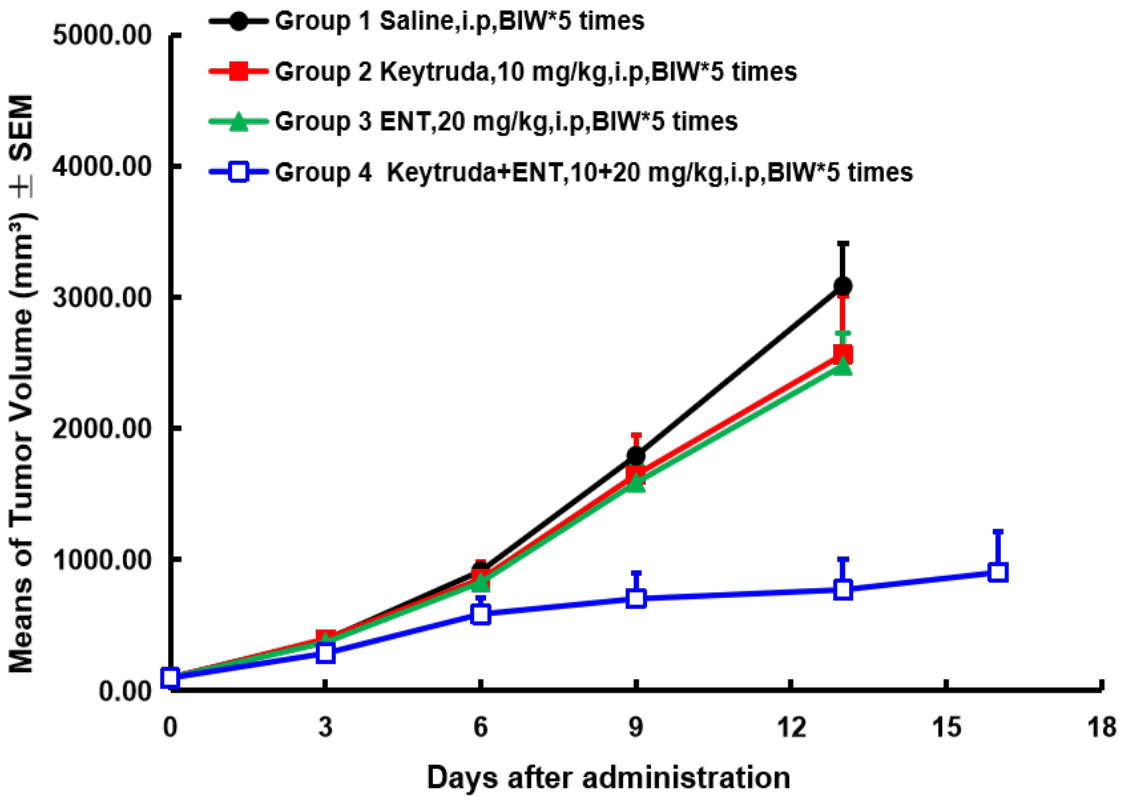


Figure 1. Expression of PD-1 in the activated spleen lymphocytes of homozygous humanized PD-1 BALB/c KI mice is detected by FACS.



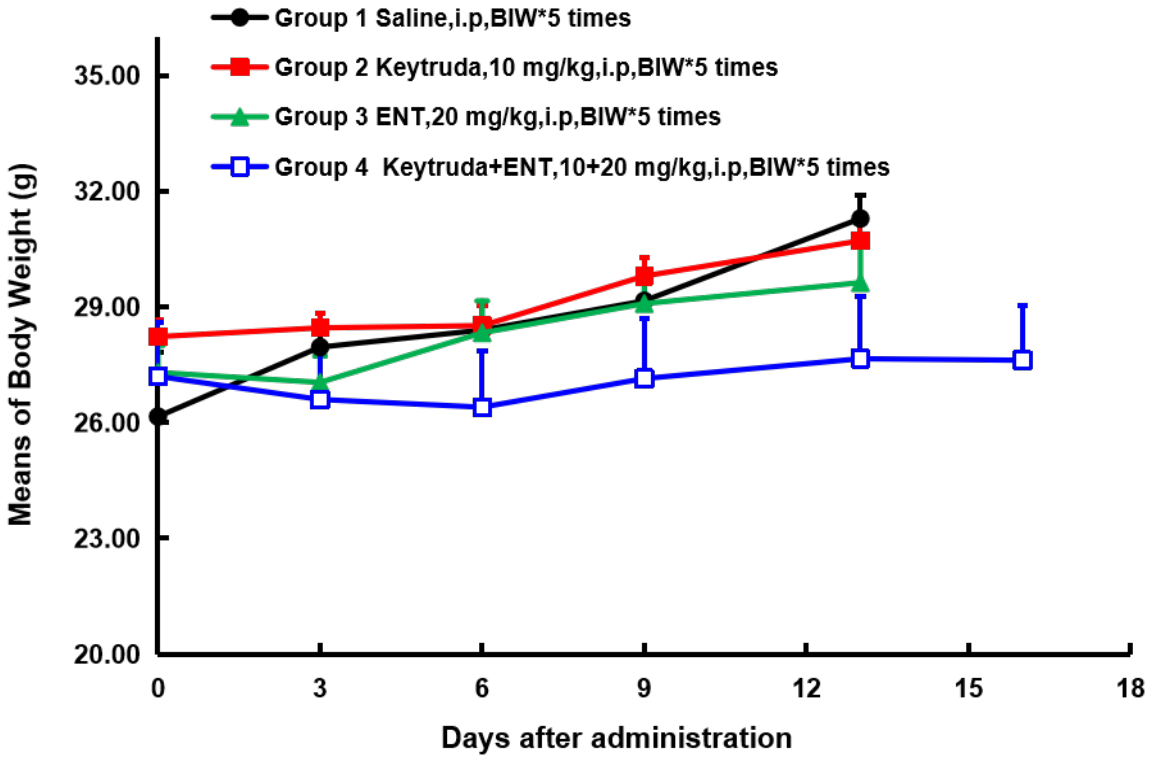
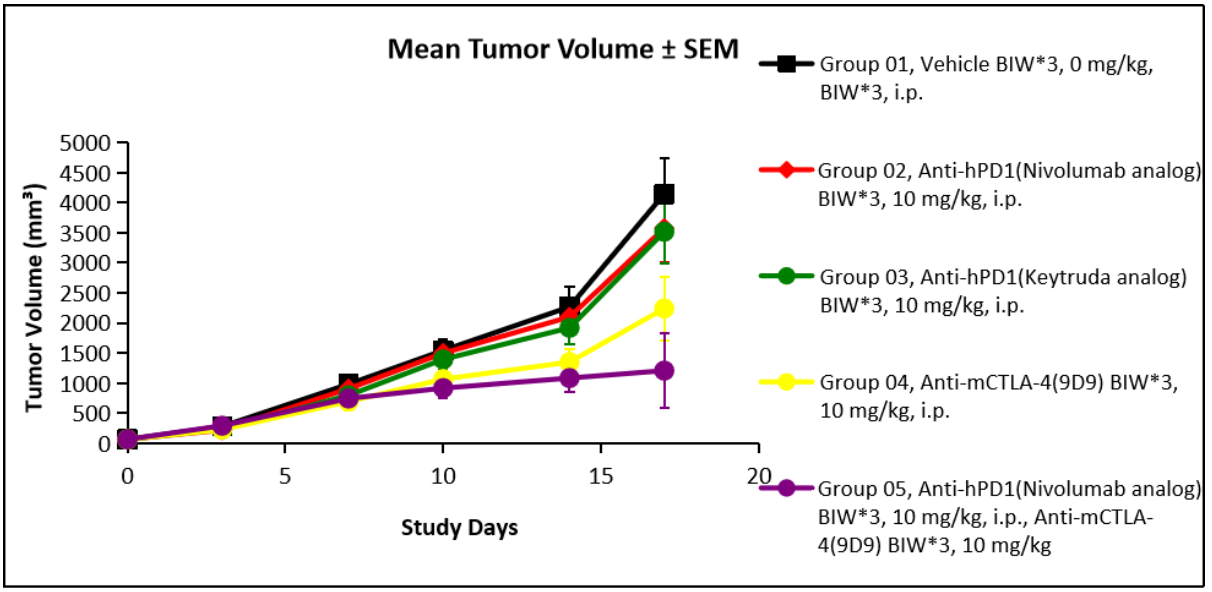


Figure 2. In vivo validation of homozygous BALB/c-hPD1 mice. The homozygous BALB/c-hPD1 mice were inoculated with CT26 cells, and randomly assigned to different groups (n=7) when the tumor grew to a volume of 100 mm3. A combinatorial treatment of anti-hPD1 antibody Keytruda and entinostat (ENT; a class I HDAC inhibitor) demonstrated a noticeable efficacy improvement compared to the same dose of single agent (A) without affecting the animal body weight (B).



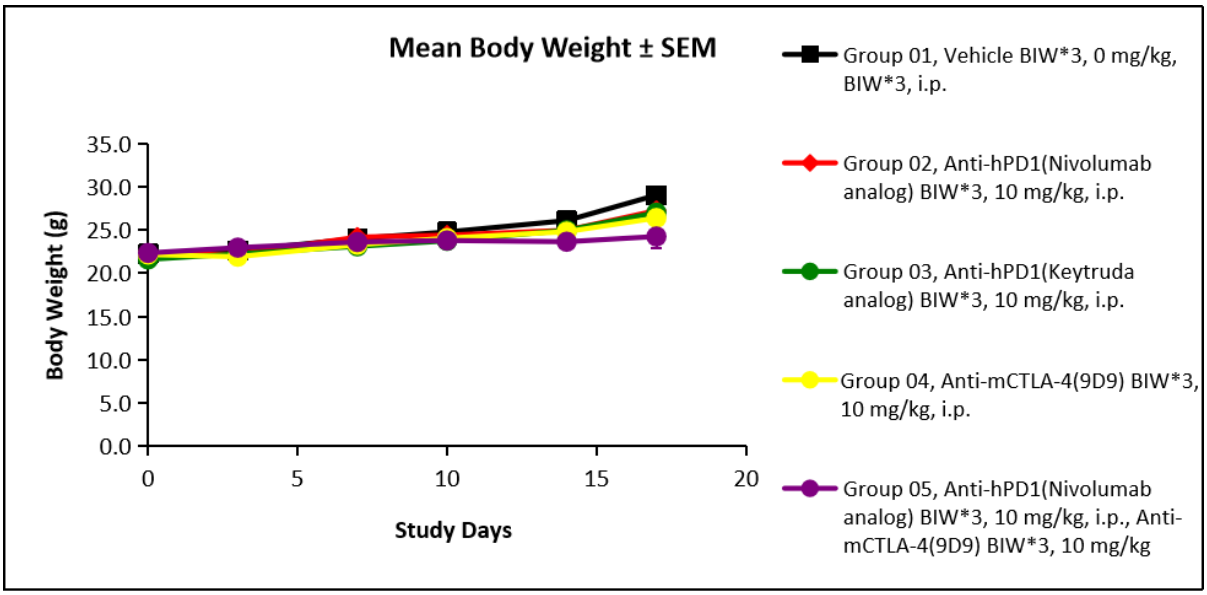


Figure 3. In vivo validation of homozygous BALB/c-hPD1 mice. The homozygous BALB/c-hPD1 mice were inoculated with CT26 cells, and randomly assigned to different groups (n=6) when the tumor grew to a volume of 65 mm^3. A combinatorial treatment of anti-hPD1 antibody Opdivo and anti-mCTLA antibody demonstrated a noticeable efficacy improvement compared to the same dose of single agent (A) without affecting the animal body weight (B). (In collaboration with Crownbio)

Humanized CD40 Mouse

Strain Name: C57BL/6-*Cd40* $^{em1(hCD40)Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00076

CD40 is a member of the tumor necrosis factor (TNF) receptor family and is also known as TNFRSF5. CD40 is expressed on antigen presenting cells (APC) such as B cells, dendritic cells (DC), and monocytes as well as many non-immune cells and various types of cancer cells. Upon binding to its ligand CD40L expressed on the antigen presenting cells (APCs), the CD40-CD40L complex plays a vital role in the helper T cells function to prime $CD8^+$ T cells. It has been demonstrated *in vitro* that agonistic CD40 mAb can improve the ability of APCs to cross-prime naive T cells to tumor antigens.

Construction strategy

The humanized CD40 mice were developed on the C57BL/6 background. The coding sequence for the extracellular domain of the mouse endogenous Cd40 was completely replaced with the human-derived sequence, resulting in the expression of a humanized, chimeric CD40 gene.

Validation data

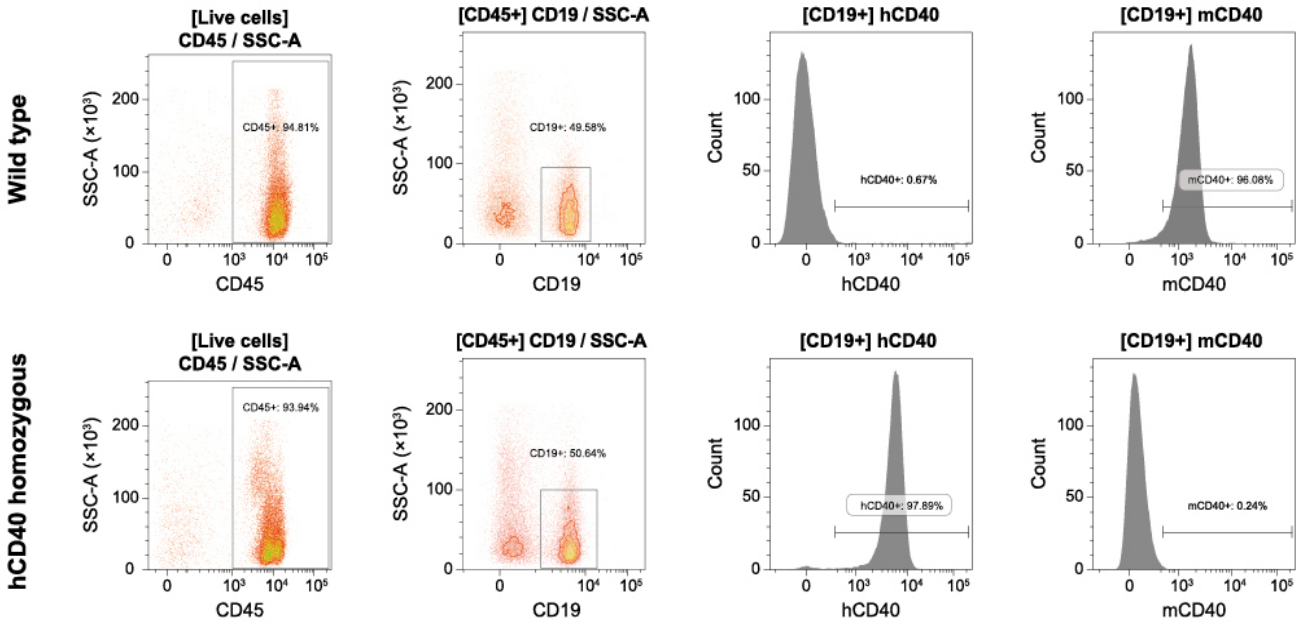


Figure 1. Detection of CD40 expression in peripheral blood cells of humanized CD40 mice. The FACS results of peripheral blood cells collected from homozygous humanized CD40 mice and wild-type mice showed that the active expression of humanized CD40 was detected in CD19 positive cells collected from homozygous humanized CD40 mice, and its expression level was similar to that of murine Cd40 expression in wild-type mice. (Completed in collaboration with CrownBio).

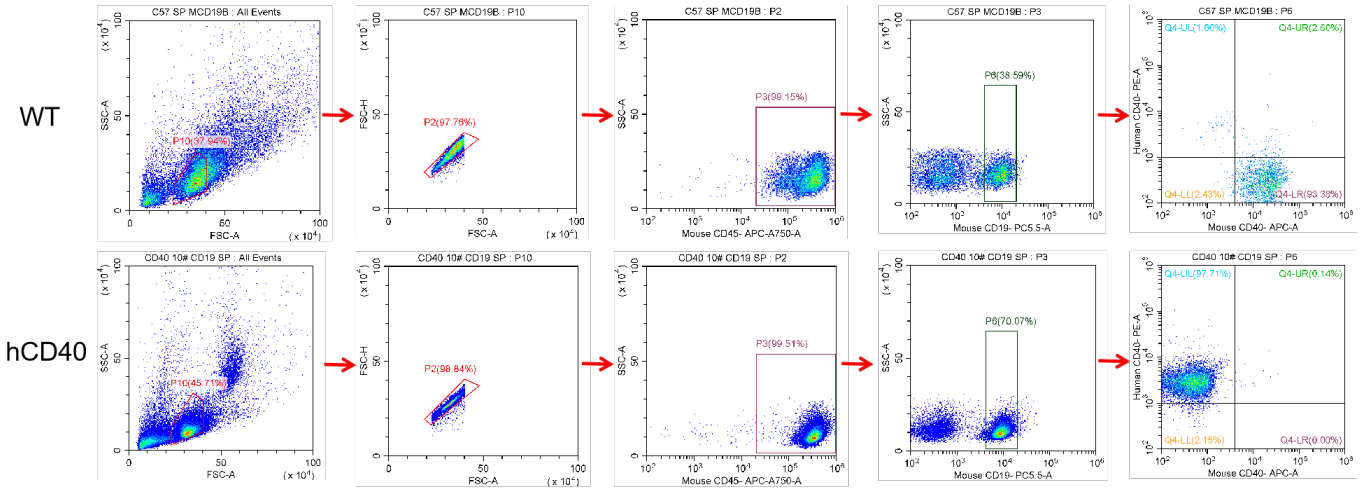
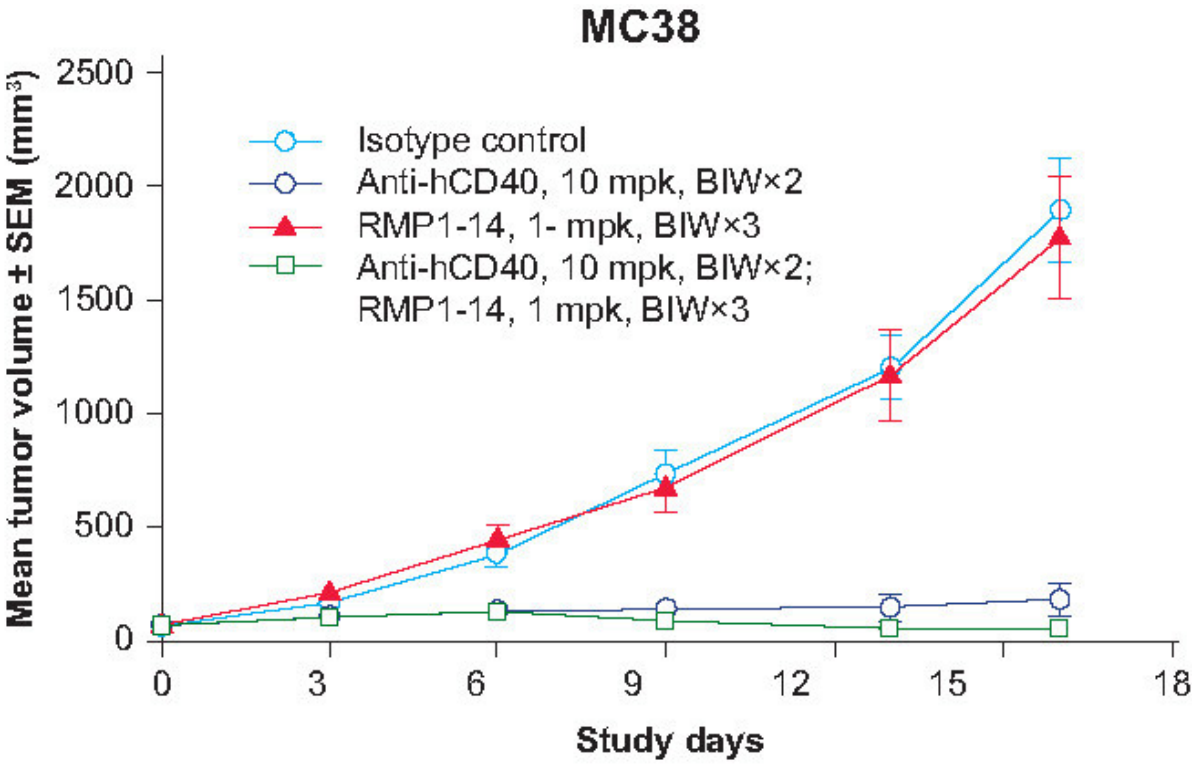


Figure 2. Analysis of human CD40 expression in the spleen by FACS. The homozygous KI animal expresses human CD40 on the CD19+ B cells.

Application

Case study 1： In vivo validation in a MC38 tumor-bearing model of humanized CTLA4 mouse



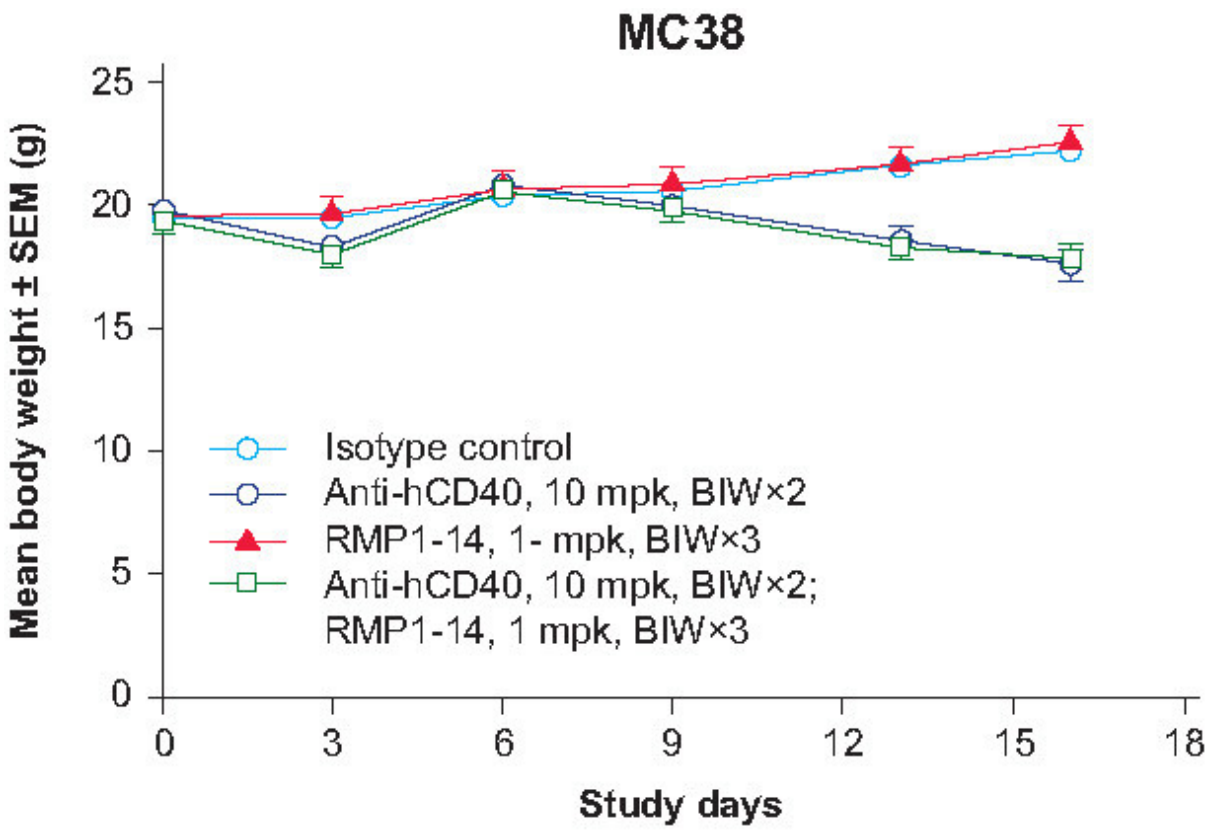
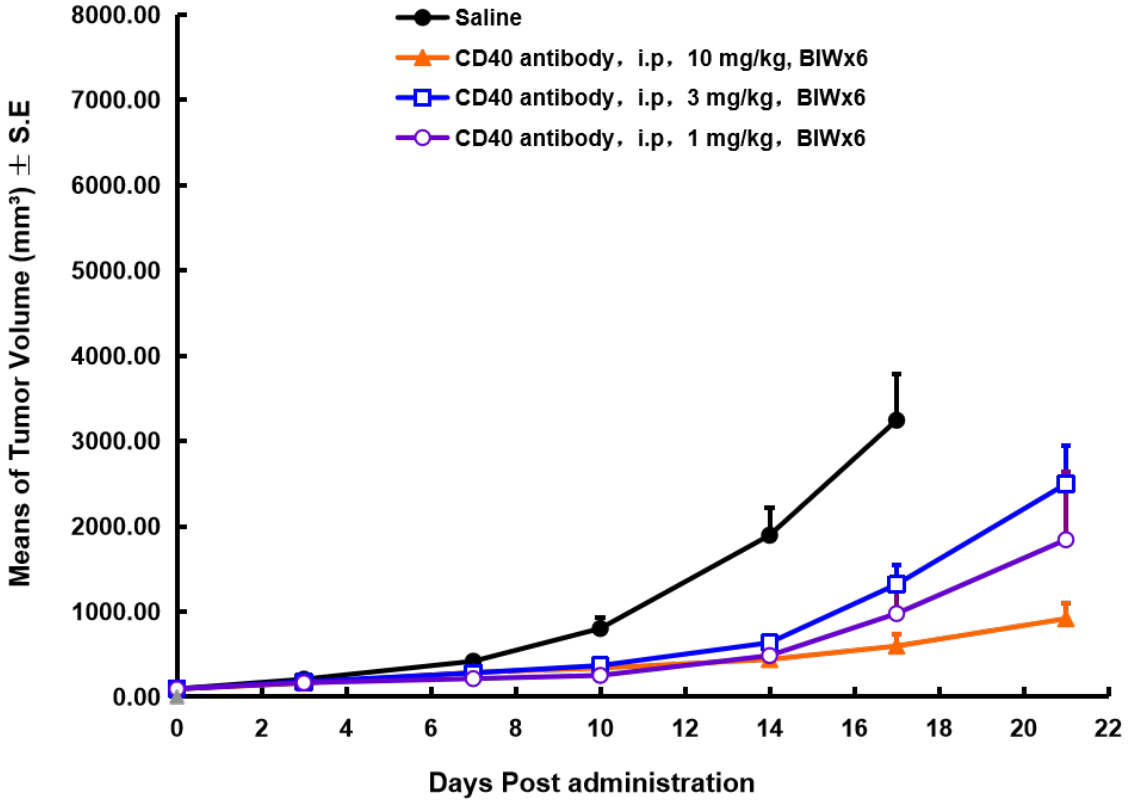


Figure 3. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized CD40 mice. Humanized CD40 mice were inoculated with MC38 colon cancer cells. After the tumors grew to 100 mm3, the animals were randomly assigned into different group (n=8). The results indicated that the antibodies targeting human CD40 showed a very significant antitumor effect ($p<0.001$). Combination of anti-CD40 and anti-PD-1 is shown more significant anti-Tumor effect. (Cooperation with CrownBio)

Case study 2: In vivo efficacy and safety evaluation of anti-human CD40 antibody using hCD40 mice

I. In vivo anti-tumor effect of an anti-human CD40 antibody in hCD40 mice.


Saline
CD40 antibody，i.p，10 mg/kg, BIWx6
CD40 antibody，i.p，3 mg/kg，BIWx6
CD40 antibody，i.p，1 mg/kg，BIWx6
Means of Tumor Volume (mm³) ± S.E
8000.00
7000.00
6000.00
5000.00
4000.00
3000.00
2000.00
1000.00
0.00
0
2
4
6
8
10
12
14
16
18
20
22
Days Post administration

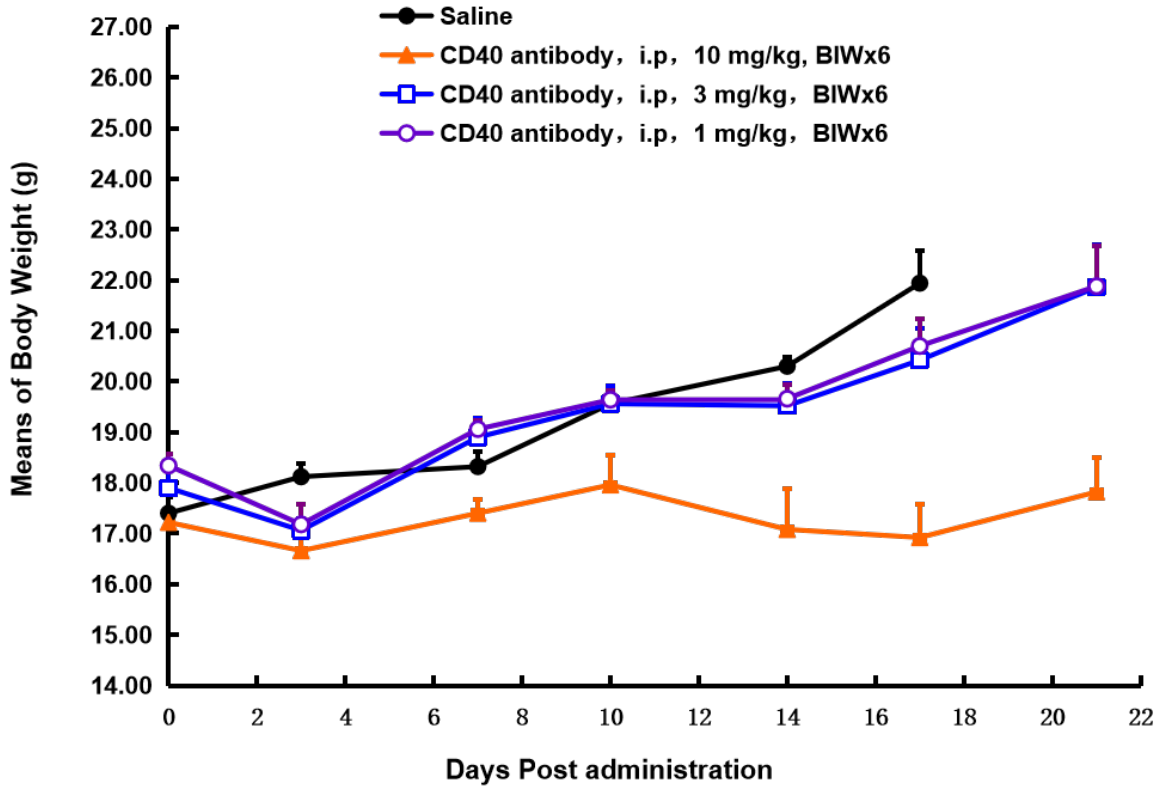
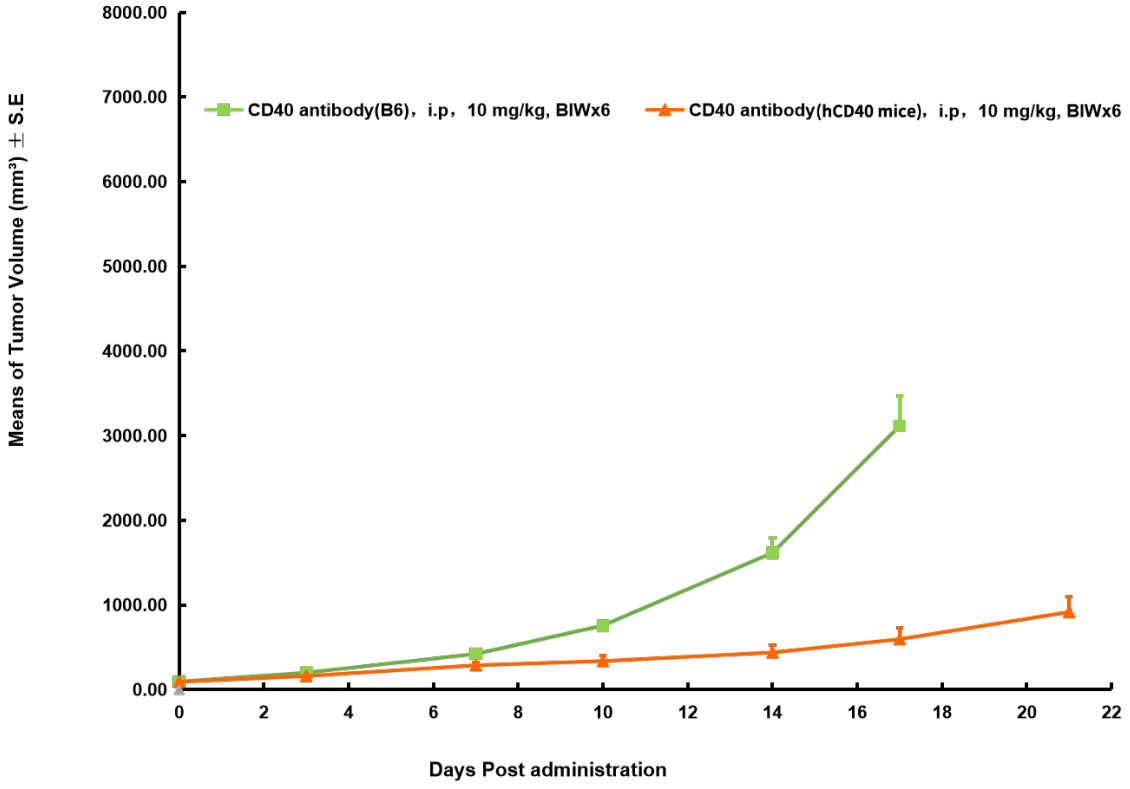

Saline
CD40 antibody，i.p，10 mg/kg, BIWx6
CD40 antibody，i.p，3 mg/kg，BIWx6
CD40 antibody，i.p，1 mg/kg，BIWx6
Means of Body Weight (g)
27.00
26.00
25.00
24.00
23.00
22.00
21.00
20.00
19.00
18.00
17.00
16.00
15.00
14.00
0
2
4
6
8
10
12
14
16
18
20
22
Days Post administration

Figure 4. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized CD40 mice. Homozygous humanized CD40 mice were inoculated with MC38 colon cancer cells. The results showed that an anti-human CD40 antibody exerted a very significant anti-tumor effect , demonstrating that the humanized CD40 mice are a good in vivo model for validating the efficacy of antibodies targeting human CD40. Mean volume ± SEM of tumor tissues(A). Mean body weight ± SEM of mice(B).

II. Evaluation of anti-human CD40 toxicity in hCD40 mice.



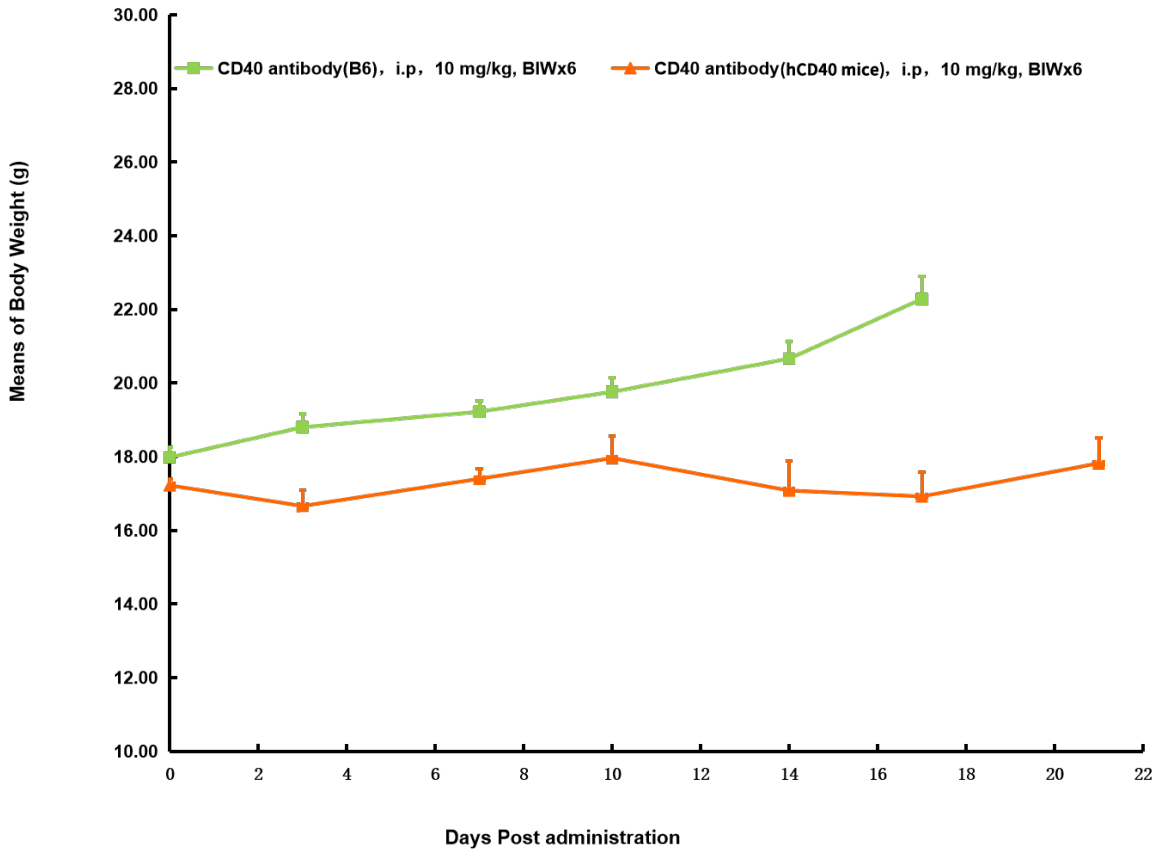
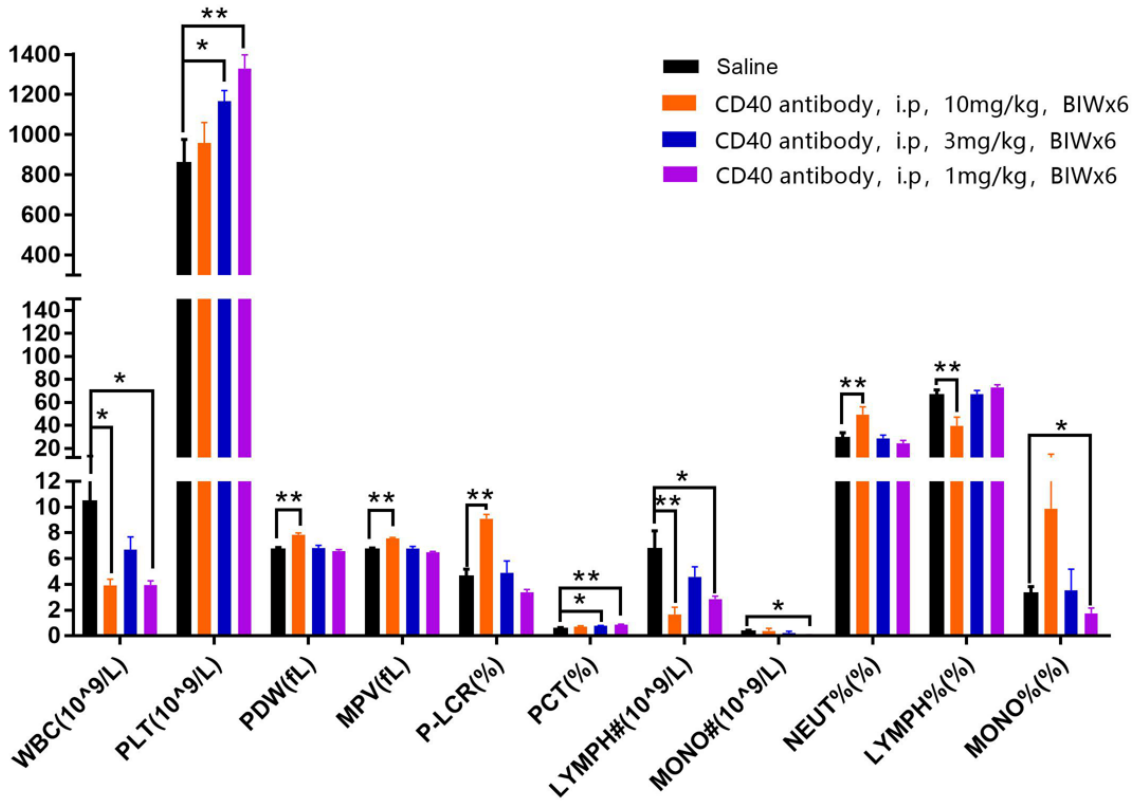
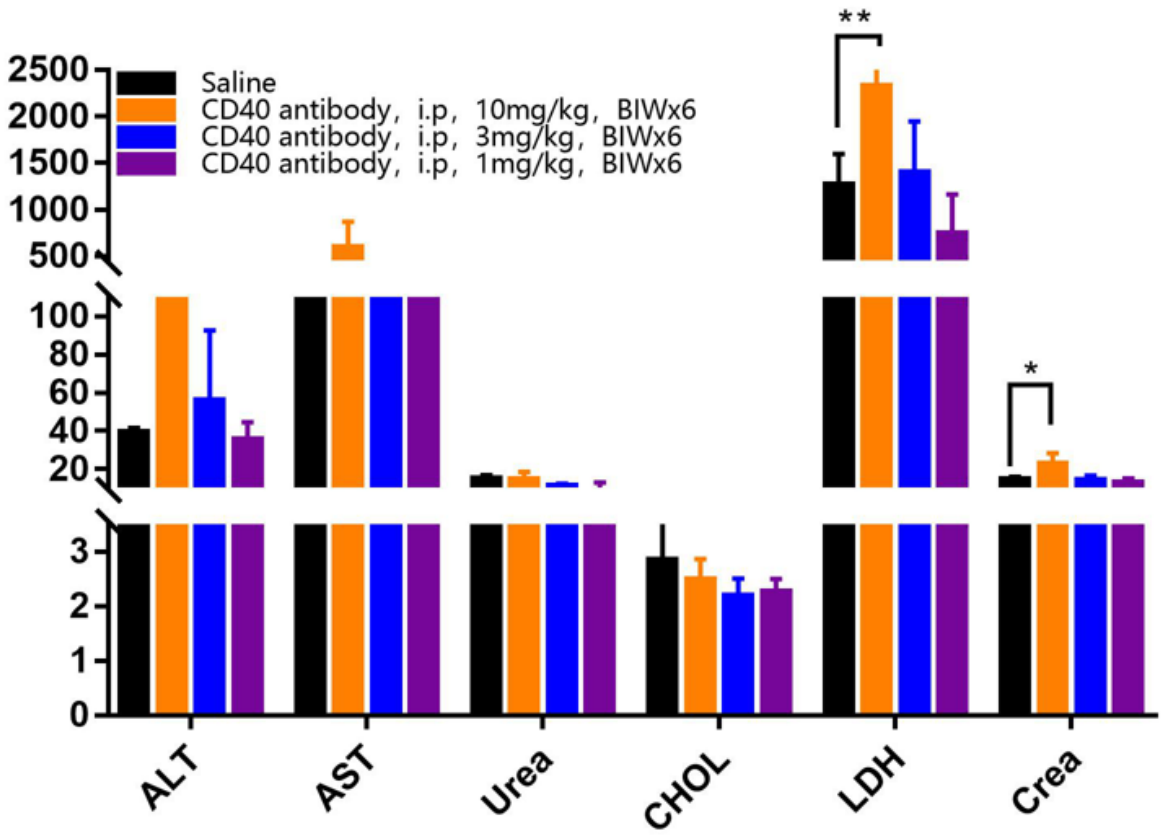


Figure 4. Tumor and weight changes over treated with anti-human CD40 antibody



Abbreviation	Stand for
WBC	White Blood Cell Count
PDW	platelet distribution width
MPV	mean platelet volume
PLCR	platelet larger cell ratio
PLT	platelet
PCT	thrombocytocrit
LYMPH#	lymphocyte count
MONO#	monocyte count
NEUT%	neutrophil count
LYMPH%	lymphocyte ratio
MONO%	monocyte ratio

Figure 5. Complete blood count (CBC) of anti-human CD40-antibody treated hCD40 mice



Abbreviation	Stand for
ALT	Alanine transaminase
AST	Aspartate transaminase
Urea	Urea
CHOL	Cholesterol
LDH	lactate dehydrogenase
Crea	creatinine

Figure 6. Blood chemistry of anti-human CD40-antibody treated hCD40 mice

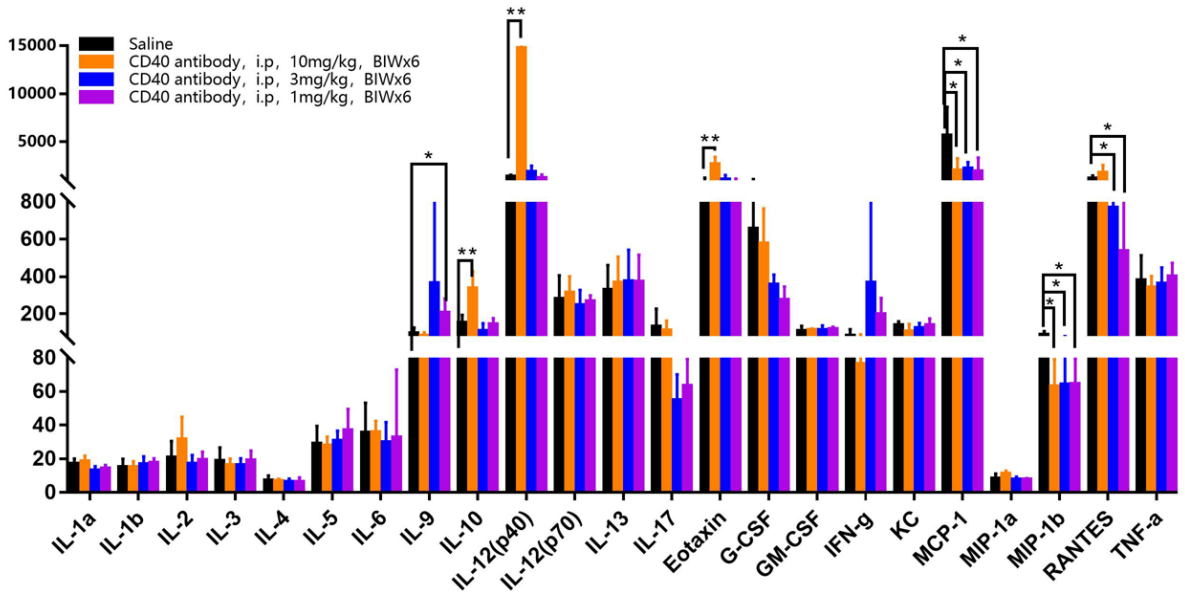


Figure 7. Cytokine analysis of anti-human CD40 antibody treated hCD40 mice.

Cytokine analysis of anti-human CD40 antibody treated MC38 tumor-bearing model of humanized CD40. Anti-human CD40 antibody treatment led to significant increase several cytokines including IL-12(p40), Eotaxin, etc.

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com** or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

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